

S2Q25

Question 25 – Stage 2

$a(= 10a_1 + a_2)$, $b(= 10b_1 + b_2)$, $c(= 10c_1 + c_2)$ and $d(= 10d_1 + d_2)$ are different integers between 01 and 99 inclusive, with $a > b$.

$$0 \leq a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2 \leq 9$$

Enter the 8-digit string $a_1a_2b_1b_2c_1c_2d_1d_2$ which produces the maximum possible integer value of $(\sqrt{a} + \sqrt{b})(\sqrt{c} - \sqrt{d})$

Working

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In [6]: from itertools import product
from math import sqrt

maximumString = ""
maximumValue = 0

f = lambda a,b,c,d: (sqrt(a) + sqrt(b)) * (sqrt(c) - sqrt(d))

for a,b,c,d in product(range(1,100),repeat=4):
    if a > b and c > d and len(set([a,b,c,d])) == len([a,b,c,d]): # Max value will be where both brackets are maximized so c>d
        currentPossible = f(a,b,c,d)
        mod = currentPossible % 1
        if min(1-mod,mod) <= 0.000000000001 and currentPossible > maximumValue:
            maximumValue = currentPossible
            maximumString = "".join([(2-len(str(i)))*"0"+str(i) for i in [a,b,c,d]])

print(f"Maximum Value: {maximumValue}")
print(f"maximum String: {maximumString}")

Maximum Value: 131.99999999999997
maximum String: 72509802
```